

Modelling and Control of Robots: Exercises

Elite Summer School - Robotics and Entrepreneurship

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Exercise 1

Simulation of Robot Manipulator



Exercise 1: Simulation of Robot Manipulator

Exercise 2: Admittance Control

Exercise 1

Simulation of Robot Manipulator - Description



The purpose of this exercise is to make a dynamical simulation of a robot. You will not derive the dynamics, but use an existing model. The exercise should be conducted in MATLAB with Robotics System Toolbox .

1. Load a UR5 robot to your workspace and visualize it (see code on next page).
2. Define a desired joint trajectory for your robot.
3. Compute the joint torques necessary to follow the desired trajectory.
4. Simulate the robot with the computed joint torque and compare the robot trajectory with the desired trajectory.

Exercise 1

Simulation of Robot Manipulator - Get Started



```
1 %% Load Robot Model
2 robot = loadrobot('universalUR5');
3 robot.DataFormat = 'column';
4 robot.Gravity = [0;0;-9.82];
5
6 %% Show Robot at Random Configuration
7 q = robot.randomConfiguration;
8 figure
9 show(robot,q)
```

Exercise 2

Admittance Control



Exercise 1: Simulation of Robot Manipulator

Exercise 2: Admittance Control

Exercise 2

Admittance Control



The purpose of this exercise is to implement an admittance control for a 6 dof robot. The exercise should be conducted in MATLAB.

1. Implement an admittance control (only translational).
2. Choose the gains of the admittance controller such that the system dynamics is critically damped.
3. Simulate the controller with a desired trajectory and a step force.
4. Extend the admittance controller to include orientation using quaternions and redo the simulations.